

NAME: _____Tovi Ding_____ SECTION DAY/TIME: _____Thu March 12th 8 AM_____

GSI: _____Idris_____ LAB PARTNER: _____Fabian Froberg, Parth Gputa_____

Lab 3: Introduction to DC circuits

Introduction

This lab introduces direct-current (DC) circuits, focusing on *conceptual understanding*. On a later worksheet, you'll integrate this qualitative understanding with mathematical problem-solving. Although everybody should do questions 1 through 4, people who already know a lot about circuits will be able to get to the challenge problems at the end.

Technical hints

- When your two batteries are hooked up in series; you can think of them as a single, double-strength battery.
- Your GSI will show you how to “transform” one circuit into another. Ask for help if you’re having trouble achieving a clean transformation.

Questions

IMPORTANT NOTE: For each question, first write your answer (prediction), then do the experiment. Finally, amend your original answer, if necessary. But don't erase your original prediction—it's helpful to have a record of what mistakes you're liable to make in the future.

1. In circuit 1, which (if either) is bigger: The current through wire A or the current through wire B? What gets “used up” when current flows through a light bulb?

Current is the same cuz it's on the same wire. Electric potential is consumed

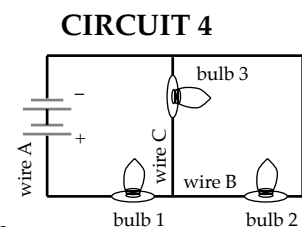
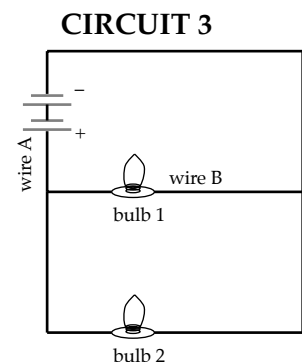
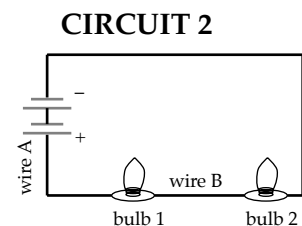
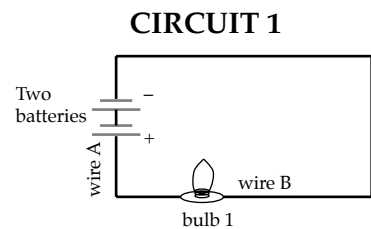
2. When circuit 1 is transformed into circuit 2 (by hooking up the 2nd light bulb), what happens to

(a) The brightness of light bulb 1?

It become dimmer becasue ther's less electric potential on the bulb

(b) The current through wire A? Why? **It is the same**

3. When circuit 1 is transformed into circuit 3, what happens to



(a) The brightness of bulb 1?

stays the same because the voltage across it is the same in circuit 3 and circuit 1, since the bulb is still directly connected to both ends of the battery

(b) The current through wire A?

same because the resistance of the bulb stays the same and the voltage is the same. making I , which is related with U and R , unchanged

(c) The current through wire B? Explain all your answers. If the experiment comes out different from your prediction, you can amend your answer by trying to explain the discrepancy.

half of current through wire A because current is evenly distributed among the 2 bulbs (assuming the 2 lightbulbs are the same)

Because the rest of the lab builds on questions 1 - 3, your GSI will go over those three questions with the whole class. If you try question 4 before this discussion, please look over your answers *after* the discussion, to take your GSI's ideas into account.

4. (*This one is hard, but very important.*) When circuit 2 is transformed into circuit 4 (by hooking up the third bulb), what happens to

(a) The current through wire A?

the current through wire 1 is larger in circuit 4. because bulb 3 and bulb 2 could be viewed as a bulb with less resistance. Thus circuit 4 connects bulb 1 and a bulb with less resistance than bulb 2 in circuit 2. So it gets more current

(b) The brightness of bulb 1?

Bulb 1's brighter in circuit 4 because $I=U/R$. I increased because there is less resistance, and so U should increase too. Thus bulb 1 gets more power and so brighter

- (c) The brightness of bulb 2? Explain your answers intuitive (*not* just with formulas). Check your answers with your GSI.

Bulb 2's dimmer. As in 4(b), there should be more voltage shared on bulb 1 because it has more resistance. This means that there is fewer voltage across bulb 3 and bulb 2. This should make bulb 2 dimmer in circuit 4 in comparison to circuit 2

5. Let I_1 denote the current through wire A in circuit 1. In terms of I_1 , what is the current through wire A in . .

(a) circuit 2? Is it $2I_1$, or $I_1/2$, or what? Explain conceptually, even if you know a formula.

$$R_{eq} = 2R, U/2R = I_1/2$$

because there are more resistance intuitively it's harder for charges to flow through so the current decreases

(b) circuit 3? Explain.

$$R_{eq} = 1/2 R, I = 2I_1$$

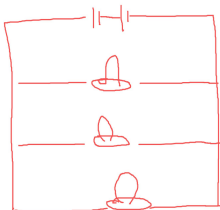
because the paths for charges flowing through the circuit expanded so there should be less resistance and so current would increase

(c) (harder) circuit 4? Explain.

$$R_{eq} = R + 1/2 R = 3/2 R, I = 2/3 I_1$$

this is intuitive because $I_{circuit 2} < I_{circuit 4} < I_{circuit 1}$
bulb 3 and bulb 2 are in parallel so it yields less resistance. bulb 1 and the combination of bulb 2 & 3 makes a resistance that is less than directly wiring 2 bulbs in series (circuit 2) but more than 1 single bulb (circuit 1)

6. With your battery, your three light bulbs, and all the wires you want, build a circuit that produces as much light as possible. Diagram the circuit here, and explain why it's the brightest.



this way we get the min resistance. cuz all bulbs are connected in parallel. and they are all directly connected to the battery. so the bulbs would generate the most power

7. With that same equipment, build a circuit that produces as *little* light as possible. Should the circuit use all three bulbs? Be sure to test this issue experimentally. Diagram your circuit, and explain why it's the dimmest.



the voltage would be distributed on the bulbs here. or that they got the most combined resistance and the current is the smallest. this makes the power the least

8. In this lab, you've built a total of six circuits: the four on page 1, the "brightest" circuit from question 6, and the "dimpest" circuit from question 7.

(a) Of those six circuits, which one has the *most* current flowing through the battery? Explain.
from question 6. because the equivalent resistance of the bulbs is the smallest and allows more current to flow through

Or since the bulbs are the brightest in 6, so that the power is the largest and because voltage stays the same, the current is larger

(b) Which has the *least* current flowing through the battery? Explain.

7. same logic as part(a). 7 has the least brightness so least power and so least current

or connecting 3 bulbs in series would yield the most resistance and allows less charges to flow through. so less current

9. Give at least *two* separate reasons why it's advantageous to wire holiday lights in parallel. (Ask your GSI if you don't know what we mean by "holiday lights.")

- the bulbs are brighter
- if one bulb broke and makes a open circuit, the other bulbs would still work

10. Are the electrical outlets in your house/room wired in series or in parallel? Explain.

parallel. because devices still work even if something is removed from the circuit. Like the fridge don't stop working if I unplug my laptop

11. In electrostatic systems, a potential difference (i.e., a voltage) always corresponds to an electric field. Is this also true about circuits? Specifically, does the potential difference between the two terminals of the battery correspond to an electric field anywhere? Or do circuits allow us to have "voltages without fields?"

there must be a field when there is voltage